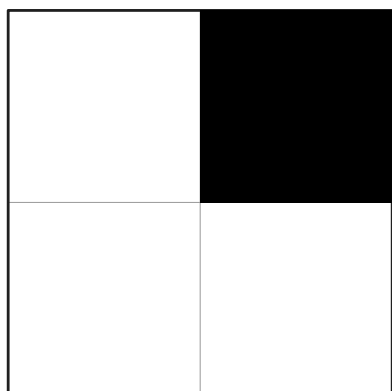
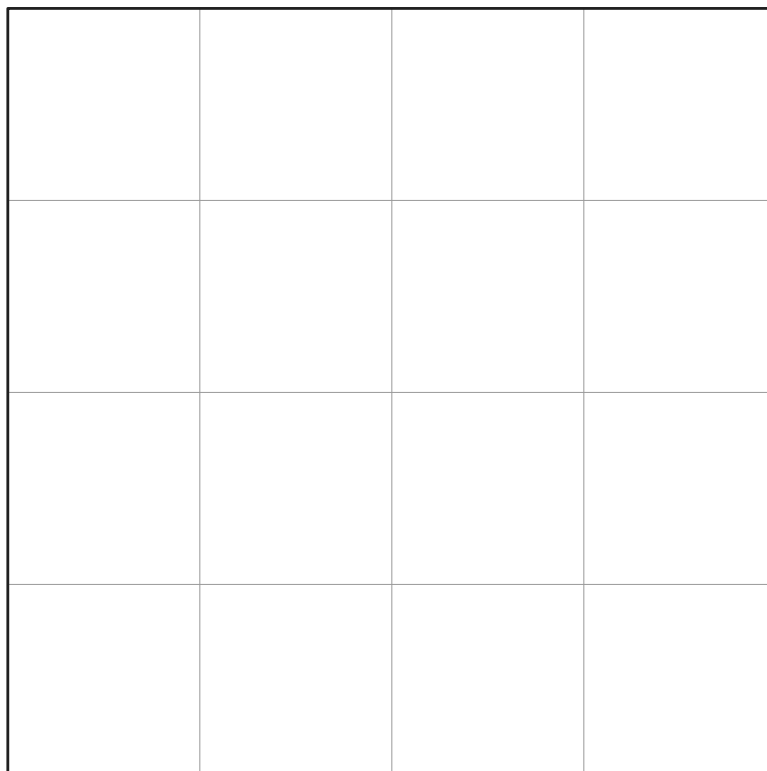


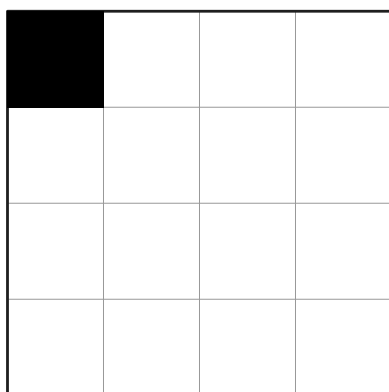
Problem 1: An L-piece  covers three squares: two squares side by side and one more square below one end. Board A is a 2-by-2 board with one black square. Cover its other three squares with one L-piece. Board B is a 4-by-4 board. Put a black square in the corner, as marked on small board 1, and cover all the other squares of board B with L-pieces. Then draw the pieces on small board 1. Do the same for the black squares marked on small boards 2 and 3.



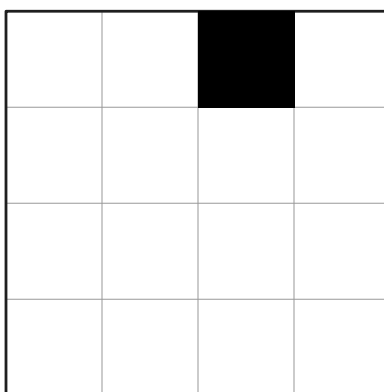
A



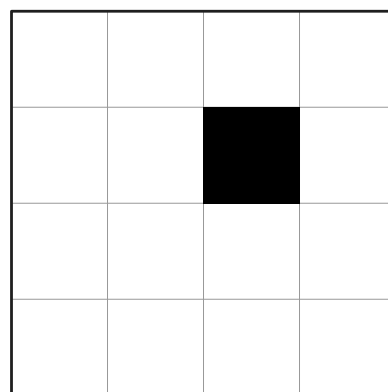
B



small board 1

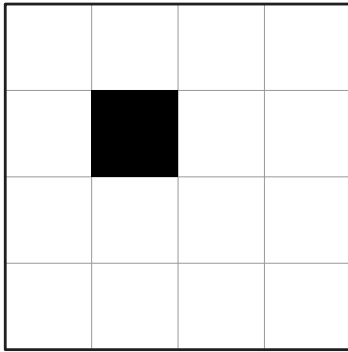


small board 2

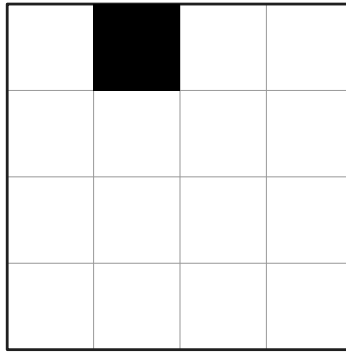


small board 3

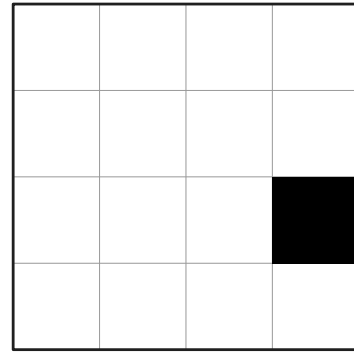
Problem 2: Put the black square on the marked square of each small board, cover the rest of board B with L-pieces, and draw the pieces on the small board. On board 6, choose the position of the black square yourself. Is there a position for which covering is impossible? Then find a method that works for every position of the black square: a way to choose the first L-piece so that whatever is left can always be finished the same way. Show your method on the two boards at the bottom by drawing only the first piece you would place. The dashed lines cut each of those boards into four 2-by-2 quarters.



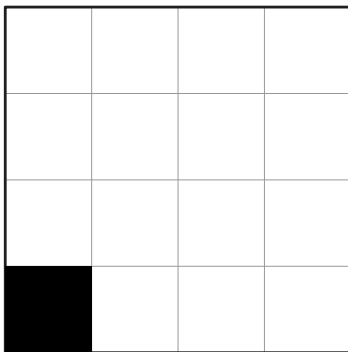
1



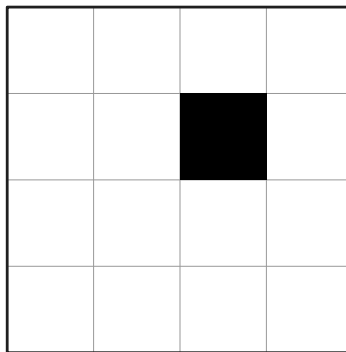
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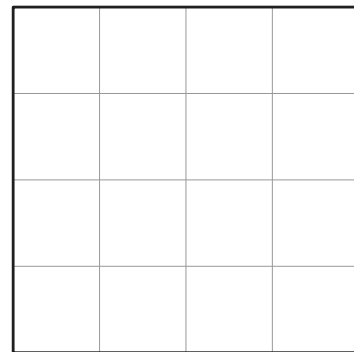
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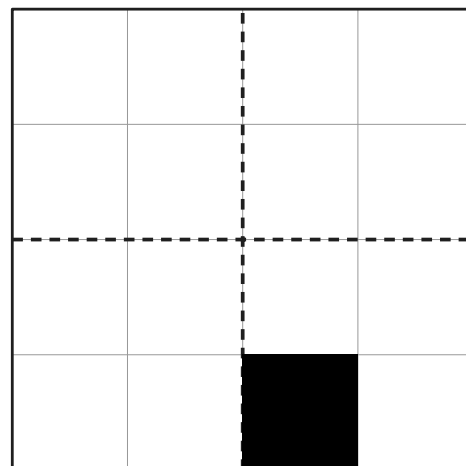
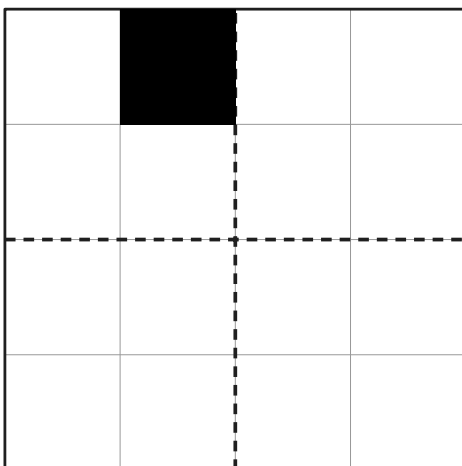
4



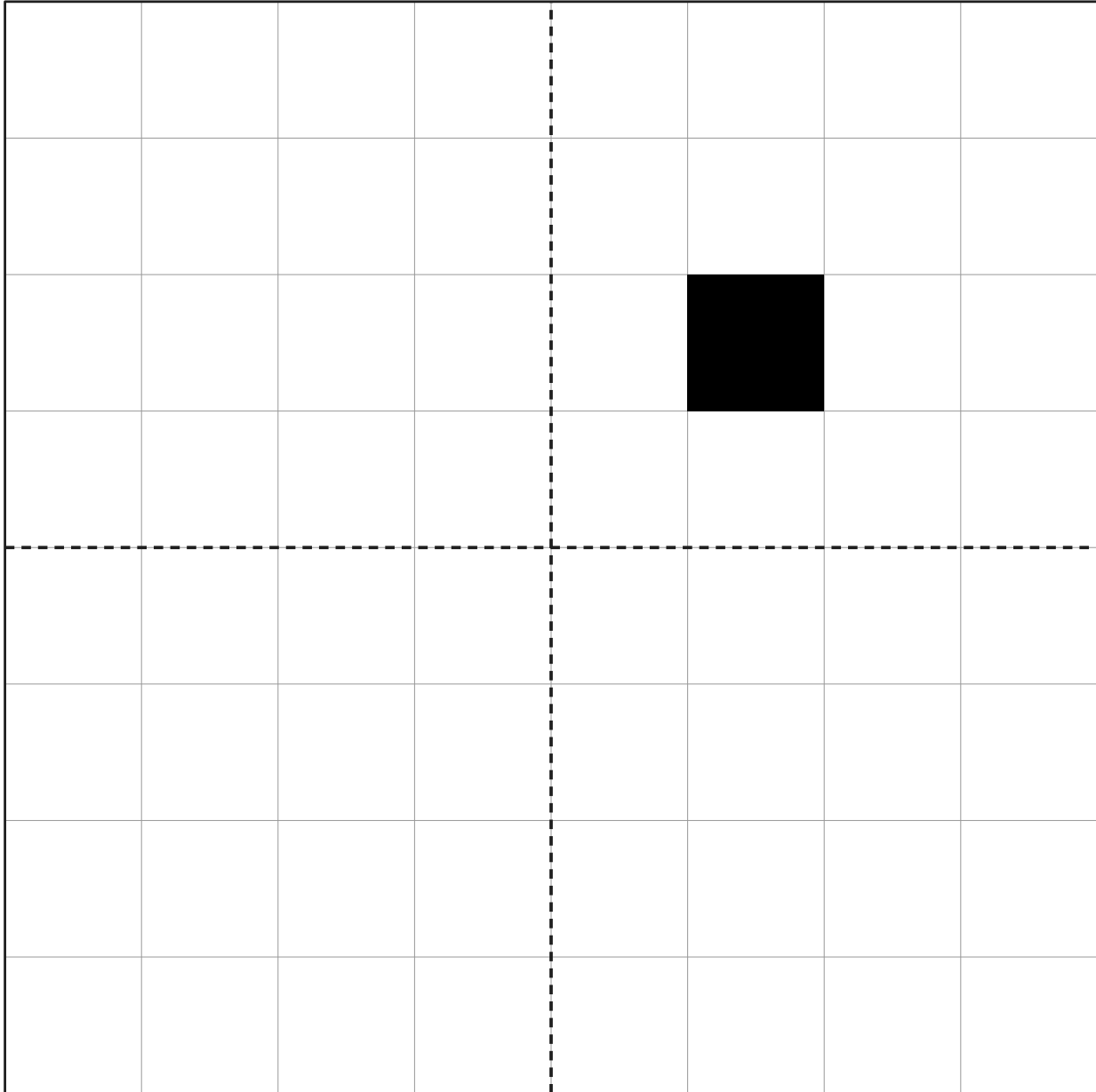
5



6

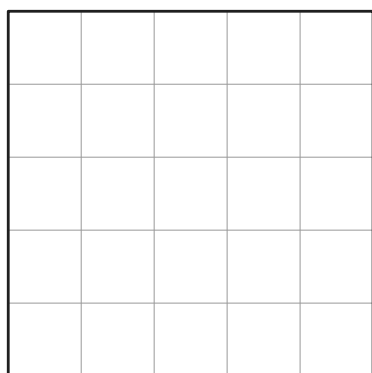
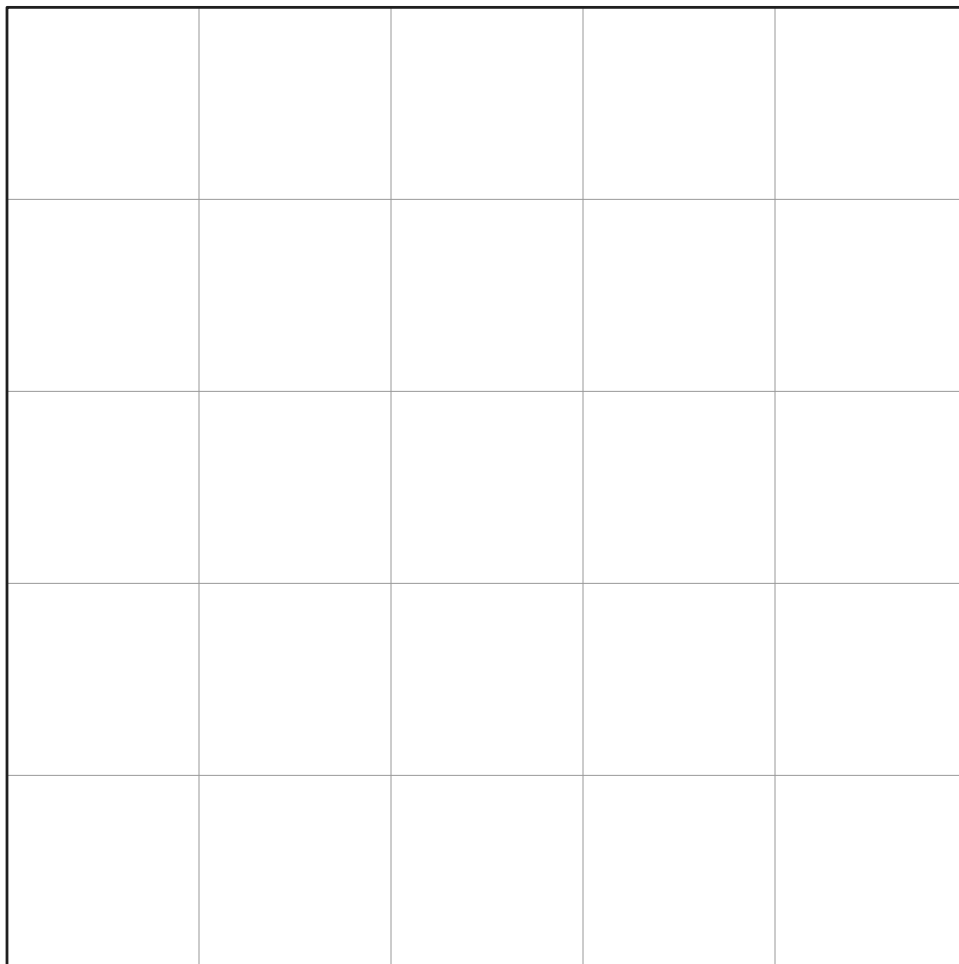


Problem 3: Cover all the squares of this 8-by-8 board except the black one with L-pieces, and draw every piece. The dashed lines cut the board into four 4-by-4 quarters; use them the way you used the quarters in Problem 2. Write how many pieces you used. Then say to the adult how you would cover a 16-by-16 board with one black square, without doing it.

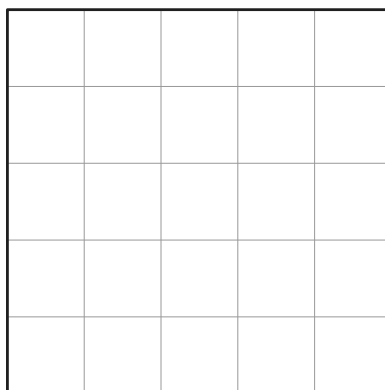


Number of L-pieces used: _____

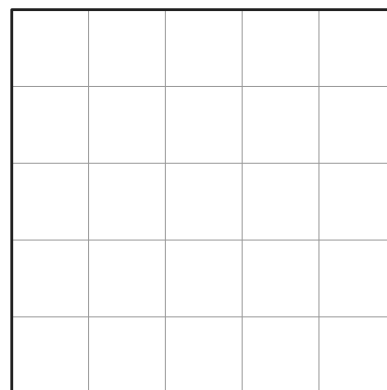
Problem 4: Put the black square on a square of this 5-by-5 board and try to cover all the other squares with L-pieces. If you can, draw a check on that square of the map; if you cannot, draw an X. Do this for every square of the map. Turning or flipping the board does not change the answer, so some squares of the map can be filled in without trying them. Draw two of the coverings you found on the small boards.



map

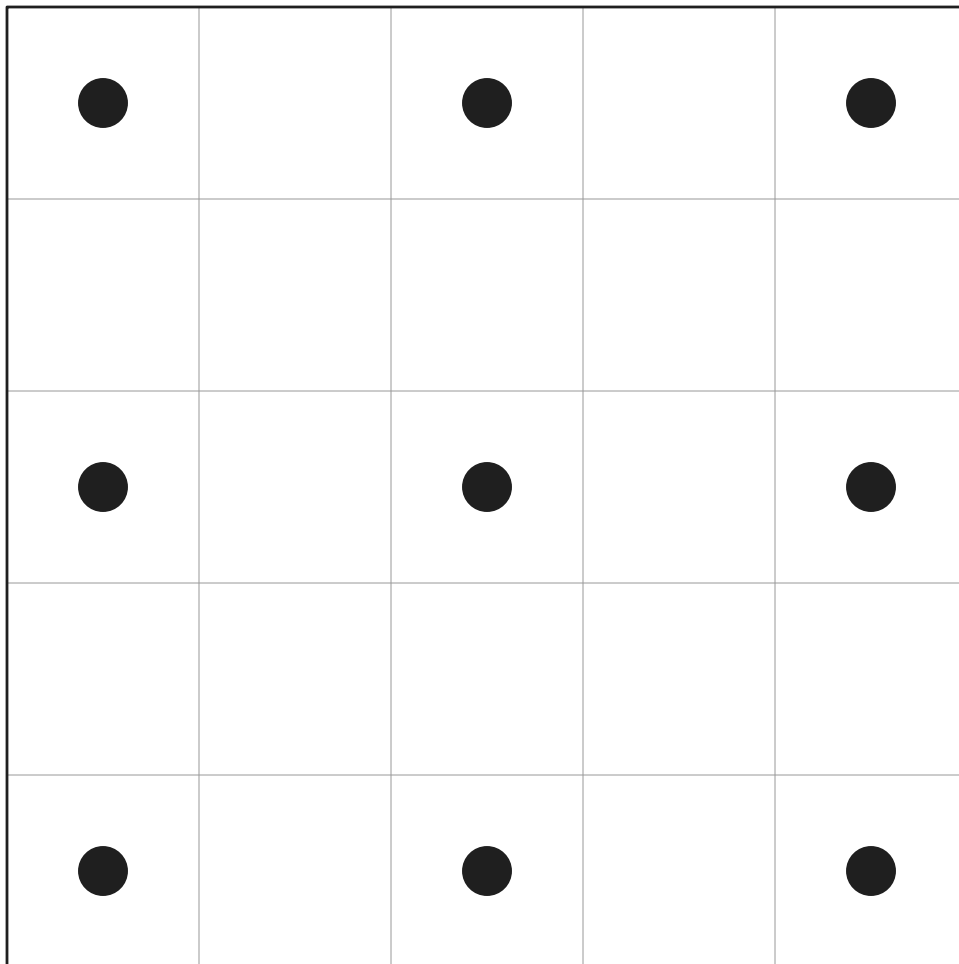


covering 1



covering 2

Problem 5: The dots mark nine squares of the board. Put one L-piece anywhere on the board and count the dots under it. Move it to a new place and count again. Do this five times and write the counts. Write how many L-pieces a covering of this board with one black square uses. Then use these numbers to say why the black square must be on a dotted square. Say it to the adult.



place 1: dots under the piece _____

place 2: dots under the piece _____

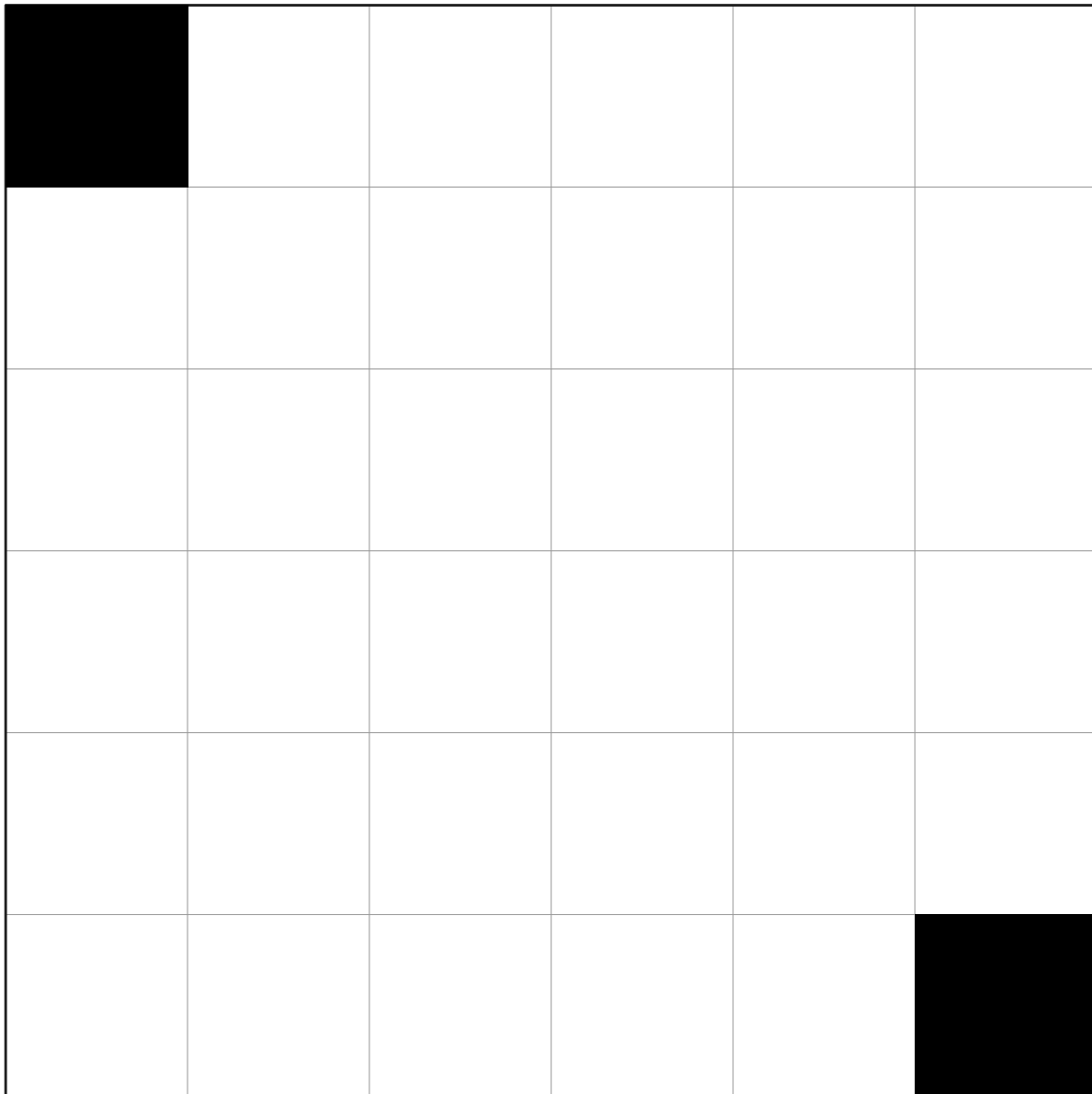
place 3: dots under the piece _____

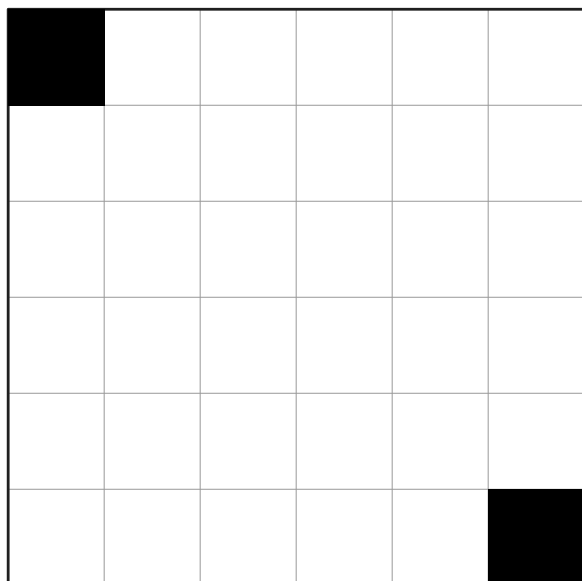
place 4: dots under the piece _____

place 5: dots under the piece _____

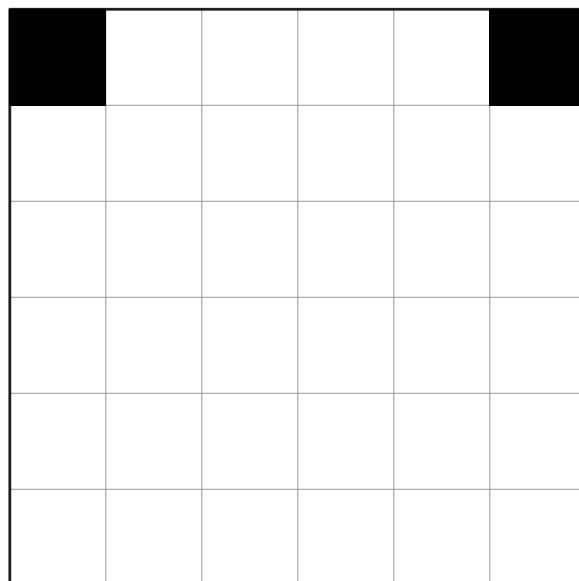
L-pieces in a covering _____

Problem 6: On this 6-by-6 board the black squares are on opposite corners. A domino covers two squares that share a side. Try to cover all the other squares with dominoes. Then find a set of squares to mark so that every domino, wherever it lies on the board, covers exactly one marked square. Shade your marked squares on the first small board on the next page, and write how many marked squares and how many unmarked squares are not black. Use the two numbers to say why covering is impossible. Then move one black square to the other corner of the same side, cover the rest with dominoes, and draw the dominoes on the second small board.



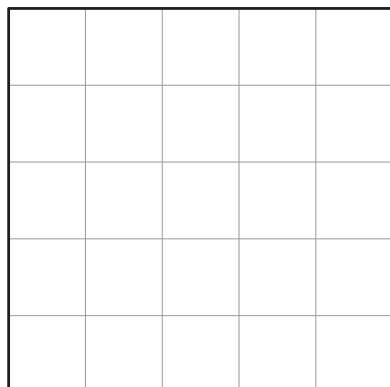


marked _____ unmarked _____

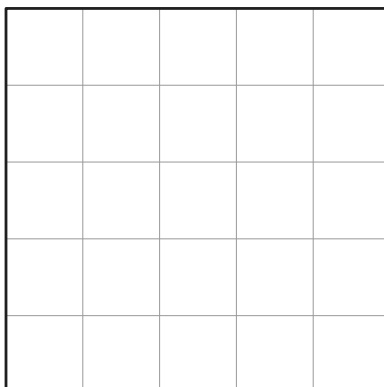


dominoes

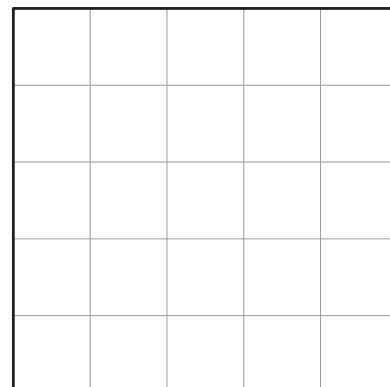
Problem 7: A straight piece  covers three squares in a row. Put the black square on a square of the 5-by-5 board from Problem 4 and try to cover all the other squares with straight pieces. For every square, record a check or an X on the map below. Then write the numbers 1, 2 and 3 on the squares of the first small board so that every straight piece, wherever it lies, covers one 1, one 2 and one 3, and count how many squares got each number. On the second small board do it in a different way. Use your counts to say which squares the black square can be on.



map

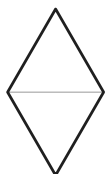


1 _____ 2 _____ 3 _____



1 _____ 2 _____ 3 _____

Problem 8: Cover the big hexagon with blue rhombi only. Record your covering on a small hexagon: color each rhombus with the color of its direction, using the key. Find three more coverings that are different from each other and from the first, and record each on its own small hexagon. Under each small hexagon write how many rhombi of each kind it has. Then say to the adult what you notice about these numbers.



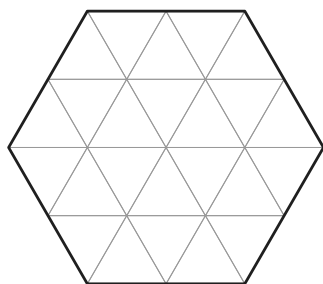
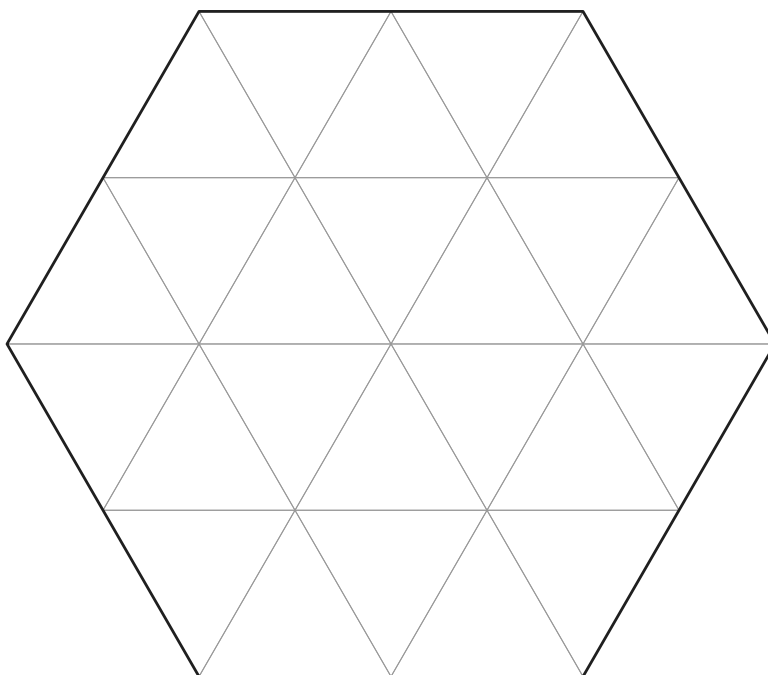
kind 1, color 1



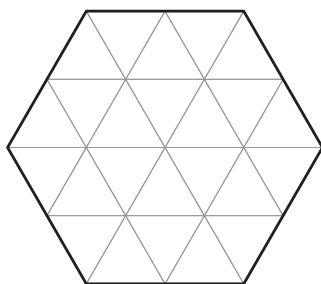
kind 2, color 2



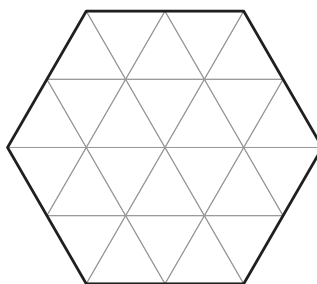
kind 3, color 3



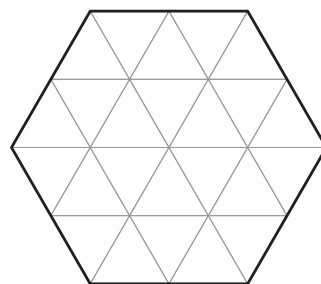
1 ___ 2 ___ 3 ___



1 ___ 2 ___ 3 ___



1 ___ 2 ___ 3 ___



1 ___ 2 ___ 3 ___